

All Waves Statistical Analyses

Marta Topor & Bertil Wegmann

January 2025

Hierarchical Bayesian Models Cross-Waves analysis

This document runs the models used to analyse the over time changes in perceptions, awareness and experience of open science practices.

The structure of the document follows the Analysis_Plans.txt file.

The variables in the datasets are explained in Meta-Data.txt.

Data Set 1 - Binary Categorical Outcomes

Crisis aware

Bayesian Logistic regression for whether participants were aware of the reproducibility crisis or not (yes/no).
The explainer variables are:

* Whether the participant was in their first or second year of study * The year the survey was conducted - 2020, 2021 and 2022

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%    97.5% n_eff
## mu             0.53     0.00 0.52   -0.51    0.22    0.53    0.85     1.56 15056
## sigma          1.37     0.00 0.45    0.73    1.05    1.30    1.60     2.44 12046
## alpha[1]       -0.30     0.00 0.15   -0.60   -0.41   -0.30   -0.20    -0.01 14093
## alpha[2]        2.13     0.01 0.78    0.84    1.59    2.05    2.58     3.87 15186
## alpha[3]        0.76     0.00 0.39    0.02    0.49    0.75    1.01     1.55 22664
## alpha[4]        1.09     0.00 0.45    0.26    0.79    1.08    1.38     2.01 20108
## alpha[5]       -0.81     0.00 0.50   -1.84   -1.14   -0.80   -0.47     0.12 20471
## alpha[6]        0.97     0.00 0.28    0.41    0.78    0.96    1.16     1.53 11498
## alpha[7]       -1.53     0.01 0.87   -3.41   -2.07   -1.47   -0.94     0.01 15877
## alpha[8]        1.97     0.00 0.27    1.45    1.79    1.97    2.15     2.51 11892
## alpha[9]        0.57     0.00 0.30   -0.02    0.36    0.57    0.77     1.17 11007
## beta1           0.92     0.00 0.16    0.62    0.82    0.92    1.03     1.23 20091
## beta2          -0.39     0.00 0.25   -0.87   -0.55   -0.39   -0.22     0.09 15614
## beta3          -0.70     0.00 0.25   -1.20   -0.86   -0.69   -0.53    -0.21 10144
## lp__          -566.25     0.03 2.80 -572.70 -567.89 -565.89 -564.23 -561.82 7517
##               Rhat
## mu                1
## sigma             1
## alpha[1]          1
## alpha[2]          1
## alpha[3]          1
```

```

## alpha[4]    1
## alpha[5]    1
## alpha[6]    1
## alpha[7]    1
## alpha[8]    1
## alpha[9]    1
## beta1       1
## beta2       1
## beta3       1
## lp__        1
##
## Samples were drawn using NUTS(diag_e) at Sat Feb 21 11:19:36 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Learnt about Open Science

Bayesian Logistic regression for whether participants learnt about open science during their studies or not (yes/no). The explainer variables are:

* Whether the participant was in their first or second year of study * The year the survey was conducted - 2020, 2021 and 2022

```

## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##          mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu        -1.18    0.00 0.39   -2.02   -1.40   -1.16   -0.93   -0.44 12758
## sigma      0.93    0.00 0.38    0.42    0.67    0.86    1.11    1.87  7882
## alpha[1]   -1.76    0.00 0.18   -2.11   -1.88   -1.76   -1.63   -1.41 12671
## alpha[2]   -0.71    0.00 0.43   -1.56   -1.00   -0.71   -0.43    0.13 20430
## alpha[3]   -1.42    0.00 0.38   -2.19   -1.66   -1.41   -1.15   -0.71 20119
## alpha[4]   -0.05    0.00 0.40   -0.81   -0.33   -0.06    0.21    0.75 16051
## alpha[5]   -2.11    0.01 0.69   -3.66   -2.52   -2.03   -1.63   -0.97 13599
## alpha[6]   -0.80    0.00 0.26   -1.31   -0.98   -0.80   -0.62   -0.29  9689
## alpha[7]   -2.21    0.01 0.94   -4.36   -2.68   -2.07   -1.58   -0.80  9781
## alpha[8]   -0.40    0.00 0.23   -0.85   -0.55   -0.40   -0.25    0.05  9266
## alpha[9]   -1.14    0.00 0.29   -1.71   -1.34   -1.14   -0.94   -0.57  9754
## beta1       0.69    0.00 0.15    0.41    0.59    0.69    0.79    0.98 17141
## beta2      -0.14    0.00 0.24   -0.61   -0.30   -0.14    0.03    0.34 12411
## beta3       0.55    0.00 0.22    0.12    0.40    0.55    0.71    1.00  8650
## lp__      -591.83    0.03 2.92  -598.54  -593.54  -591.49  -589.73  -587.16  7018
##          Rhat
## mu          1
## sigma        1
## alpha[1]     1
## alpha[2]     1
## alpha[3]     1
## alpha[4]     1
## alpha[5]     1
## alpha[6]     1
## alpha[7]     1
## alpha[8]     1
## alpha[9]     1

```

```

## beta1      1
## beta2      1
## beta3      1
## lp__       1
##
## Samples were drawn using NUTS(diag_e) at Sat Feb 21 11:21:05 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Can Explain Open Science

Bayesian Logistic regression for whether participants can explain open science or not (yes/no). The explainer variables are:

* Whether the participant was in their first or second year of study * The year the survey was conducted - 2020, 2021 and 2022

```

## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##          mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu        -1.02    0.00  0.20   -1.40   -1.15   -1.02   -0.90   -0.62  4651
## sigma      0.32    0.00  0.18    0.08    0.19    0.28    0.40    0.75  2884
## alpha[1]   -1.29    0.00  0.16   -1.60   -1.39   -1.28   -1.18   -0.98  6904
## alpha[2]   -1.14    0.00  0.31   -1.81   -1.31   -1.12   -0.94   -0.56 10884
## alpha[3]   -1.08    0.00  0.26   -1.63   -1.25   -1.08   -0.91   -0.57 10514
## alpha[4]   -0.70    0.01  0.35   -1.28   -0.94   -0.73   -0.47    0.06  3073
## alpha[5]   -1.17    0.00  0.32   -1.89   -1.34   -1.15   -0.96   -0.60  9655
## alpha[6]   -0.90    0.00  0.23   -1.34   -1.06   -0.91   -0.75   -0.44  3078
## alpha[7]   -1.05    0.00  0.35   -1.79   -1.26   -1.05   -0.84   -0.35 10133
## alpha[8]   -0.88    0.00  0.21   -1.28   -1.02   -0.88   -0.73   -0.47  2944
## alpha[9]   -0.99    0.00  0.24   -1.44   -1.16   -1.00   -0.84   -0.51  4327
## beta1      0.36    0.00  0.14    0.09    0.27    0.36    0.46    0.63 10896
## beta2      0.19    0.00  0.22   -0.25    0.04    0.19    0.34    0.63  4314
## beta3      0.56    0.00  0.20    0.17    0.43    0.56    0.70    0.96  2882
## lp__     -614.99    0.10  4.02  -623.19 -617.52 -614.90 -612.32 -607.18 1688
##          Rhat
## mu          1
## sigma       1
## alpha[1]    1
## alpha[2]    1
## alpha[3]    1
## alpha[4]    1
## alpha[5]    1
## alpha[6]    1
## alpha[7]    1
## alpha[8]    1
## alpha[9]    1
## beta1       1
## beta2       1
## beta3       1
## lp__        1
##
## Samples were drawn using NUTS(diag_e) at Sat Feb 21 11:23:02 2026.

```

```
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Applicability

Bayesian Logistic regression for whether participants think that the issues discussed in the questionnaire are relevant to them (yes/no). The explainer variables are: * Whether the participant was in their first or second year of study * The year the survey was conducted - 2020, 2021 and 2022

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##           mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu          1.99     0.00 0.25    1.54    1.82    1.98    2.14    2.52 2888
## sigma       0.31     0.01 0.23    0.05    0.15    0.26    0.42    0.89 1751
## alpha[1]    1.79     0.00 0.20    1.41    1.65    1.79    1.92    2.18 4157
## alpha[2]    1.81     0.00 0.36    0.99    1.61    1.83    2.04    2.48 6534
## alpha[3]    2.10     0.01 0.38    1.49    1.85    2.05    2.31    3.00 3486
## alpha[4]    2.17     0.01 0.43    1.52    1.88    2.10    2.39    3.23 2864
## alpha[5]    1.83     0.00 0.36    1.04    1.63    1.84    2.05    2.50 7036
## alpha[6]    2.17     0.01 0.32    1.61    1.94    2.14    2.37    2.86 1814
## alpha[7]    1.94     0.00 0.41    1.11    1.71    1.93    2.16    2.79 6839
## alpha[8]    2.03     0.01 0.27    1.54    1.84    2.02    2.20    2.60 2567
## alpha[9]    2.07     0.01 0.31    1.52    1.85    2.05    2.26    2.75 2538
## beta1       0.75     0.00 0.20    0.36    0.61    0.75    0.88    1.15 6601
## beta2      -0.46     0.00 0.30   -1.05   -0.67   -0.46   -0.26    0.12 4561
## beta3      -0.83     0.01 0.27   -1.39   -1.01   -0.82   -0.64   -0.32 2596
## lp__      -397.02     0.18 5.61  -407.23  -400.89  -397.36  -393.51  -385.54 1013
##           Rhat
## mu          1.00
## sigma       1.00
## alpha[1]    1.00
## alpha[2]    1.00
## alpha[3]    1.00
## alpha[4]    1.00
## alpha[5]    1.00
## alpha[6]    1.00
## alpha[7]    1.00
## alpha[8]    1.00
## alpha[9]    1.00
## beta1       1.00
## beta2       1.00
## beta3       1.00
## lp__        1.01
##
## Samples were drawn using NUTS(diag_e) at Sat Feb 21 11:26:39 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Data Set 2 - Truncated Poisson Regression

Awarweness Score Total

Bayesian regression for how much awareness of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

* Whether the participant was in their first or second year of study * The year the survey was conducted - 2020, 2021 and 2022

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu             1.54     0.00 0.06    1.42    1.50    1.53    1.57    1.66  3752
## sigma          0.10     0.00 0.07    0.01    0.05    0.09    0.14    0.28   694
## alpha[1]       1.52     0.00 0.04    1.45    1.50    1.52    1.55    1.60  2537
## alpha[2]       1.53     0.00 0.08    1.37    1.49    1.53    1.58    1.70  7148
## alpha[3]       1.51     0.00 0.07    1.37    1.47    1.51    1.55    1.64  4846
## alpha[4]       1.66     0.00 0.11    1.49    1.57    1.65    1.74    1.90   727
## alpha[5]       1.46     0.00 0.09    1.25    1.40    1.47    1.52    1.60  2292
## alpha[6]       1.56     0.00 0.09    1.40    1.50    1.55    1.61    1.77  3792
## alpha[7]       1.52     0.00 0.10    1.29    1.46    1.52    1.57    1.72  7295
## alpha[8]       1.51     0.00 0.06    1.40    1.48    1.52    1.55    1.63  4151
## alpha[9]       1.56     0.00 0.13    1.32    1.49    1.54    1.62    1.88  3710
## beta1          0.14     0.00 0.05    0.04    0.10    0.14    0.17    0.23  1684
## beta2          0.07     0.00 0.06   -0.04    0.03    0.07    0.11    0.18   998
## beta3          4.41     0.01 0.32    3.74    4.19    4.42    4.64    4.93  1462
## lp__          3914.31    0.33 5.55 3904.80 3910.64 3913.60 3917.32 3927.74   289
##               Rhat
## mu             1.00
## sigma          1.01
## alpha[1]       1.00
## alpha[2]       1.00
## alpha[3]       1.00
## alpha[4]       1.01
## alpha[5]       1.00
## alpha[6]       1.00
## alpha[7]       1.00
## alpha[8]       1.00
## alpha[9]       1.00
## beta1          1.00
## beta2          1.00
## beta3          1.00
## lp__          1.02
##
## Samples were drawn using NUTS(diag_e) at Sat Feb 21 11:34:05 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 767

Year 1 N = 424

Year 2 N = 343
 2020 N = 340
 2021 N = 184
 2022 N = 243

Var1	Freq
0	12
1	30
2	41
3	71
4	97
5	85
6	78
7	55
8	298

The output shows that beta credibility intervals are significant for Beta 1.

2.5% = 0.0386924

97.5% = 0.2310151

The interpretation is that students across all waves, who were in their second year of study were more likely to have higher awareness of open science practices compared to students in the first year of study.

The output shows that beta credibility intervals are significant for Beta 3.

2.5% = 3.7434138

97.5% = 4.930699

The interpretation is that students who answered the survey in 2022 were significantly more likely to have higher awareness of open science practices compared to students who answered the survey in 2020.

177 divergent transitions.

Experience Score Total

Bayesian regression for how much experience of open and reproducible research the students had - scores range from 0 to 8.

The explainer variables are:

* Whether the participant was in their first or second year of study * The year the survey was conducted - 2020, 2021 and 2022

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd    2.5%    25%    50%    75%   97.5% n_eff
## mu           0.74    0.00 0.07    0.58    0.69    0.74    0.78    0.87  8141
## sigma        0.14    0.00 0.07    0.05    0.09    0.13    0.18    0.33  5224
## alpha[1]     0.73    0.00 0.05    0.64    0.70    0.73    0.76    0.82 10143
## alpha[2]     0.72    0.00 0.11    0.50    0.66    0.73    0.79    0.93 15926
## alpha[3]     0.73    0.00 0.09    0.55    0.67    0.73    0.79    0.90 16963
## alpha[4]     0.82    0.00 0.10    0.62    0.75    0.81    0.88    1.03 13634
## alpha[5]     0.65    0.00 0.12    0.39    0.58    0.66    0.73    0.86 11625
## alpha[6]     0.79    0.00 0.07    0.65    0.74    0.79    0.84    0.93  8459
## alpha[7]     0.60    0.00 0.17    0.19    0.51    0.63    0.72    0.86  8282
## alpha[8]     0.90    0.00 0.07    0.77    0.86    0.90    0.95    1.03  6736
```

```

## alpha[9]    0.67    0.00 0.08    0.51    0.62    0.67    0.72    0.82 8666
## beta1       0.26    0.00 0.04    0.17    0.23    0.26    0.29    0.34 12563
## beta2      -0.39    0.00 0.07   -0.53   -0.43   -0.39   -0.34   -0.25 9816
## beta3      -0.02    0.00 0.06   -0.14   -0.06   -0.02    0.02    0.10 7725
## lp__       -359.81    0.05 3.33 -367.13 -361.88 -359.52 -357.45 -354.05 4657
##           Rhat
## mu          1
## sigma       1
## alpha[1]    1
## alpha[2]    1
## alpha[3]    1
## alpha[4]    1
## alpha[5]    1
## alpha[6]    1
## alpha[7]    1
## alpha[8]    1
## alpha[9]    1
## beta1       1
## beta2       1
## beta3       1
## lp__        1
##
## Samples were drawn using NUTS(diag_e) at Sat Feb 21 11:38:43 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).

```

Number of participants in the analysis

Total N = 987
 Year 1 N = 571
 Year 2 N = 416
 2020 N = 340
 2021 N = 184
 2022 N = 463

Var1	Freq
0	211
1	167
2	186
3	178
4	126
5	75
6	28
7	11
8	5

The output shows that beta credibility intervals are significant for Beta 1.

2.5% = 0.1736855

97.5% = 0.3419942

The interpretation is that students across all waves, who were in their second year of study were more likely to have higher experience of open science practices compared to students in the first year of study.

The output shows that beta credibility intervals are significant for Beta 2 in the negative direction.

2.5% = -0.5282544
 97.5% = -0.2451832

The interpretation is that students who answered the survey in 2020 were significantly more likely to have higher experience of open science practices compared to students who answered the survey in 2022.

Data Set 2 - Linear Regression

Conceptual Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research. The scores are between -2 and 2 but will be changed by adding 2.001. This way the scale will be from 0 to 4.

The explainer variables are:

* Whether the participant was in their first or second year of study * The year the survey was conducted - 2020, 2021 and 2022

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##               mean se_mean   sd  2.5%   25%   50%   75%  97.5% n_eff Rhat
## mu           0.59    0.00 0.05  0.49  0.56  0.59  0.62  0.69 14141  1
## sigma        0.38    0.00 0.01  0.36  0.38  0.38  0.39  0.40 22051  1
## sigma_a      0.12    0.00 0.05  0.05  0.08  0.11  0.14  0.23 11571  1
## alpha[1]     0.55    0.00 0.03  0.50  0.53  0.55  0.57  0.60 14343  1
## alpha[2]     0.53    0.00 0.07  0.38  0.48  0.53  0.58  0.66 20110  1
## alpha[3]     0.62    0.00 0.06  0.51  0.58  0.61  0.65  0.73 21396  1
## alpha[4]     0.70    0.00 0.07  0.57  0.65  0.70  0.75  0.84 18633  1
## alpha[5]     0.49    0.00 0.07  0.34  0.44  0.49  0.54  0.63 17713  1
## alpha[6]     0.70    0.00 0.05  0.61  0.67  0.70  0.73  0.79 10985  1
## alpha[7]     0.58    0.00 0.09  0.40  0.52  0.59  0.64  0.77 19310  1
## alpha[8]     0.66    0.00 0.04  0.59  0.63  0.66  0.69  0.74 10742  1
## alpha[9]     0.50    0.00 0.05  0.41  0.47  0.50  0.54  0.60 10889  1
## beta1        0.05    0.00 0.03  0.00  0.03  0.05  0.06  0.09 18060  1
## beta2       -0.07    0.00 0.04 -0.14 -0.09 -0.07 -0.04  0.01 14049  1
## beta3        0.01    0.00 0.04 -0.06 -0.01  0.01  0.04  0.09  9875  1
## lp__        471.23    0.04 2.97 464.51 469.43 471.56 473.37 476.04  7055  1
##
## Samples were drawn using NUTS(diag_e) at Sat Feb 21 11:44:12 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 983
 Year 1 N = 569
 Year 2 N = 414
 2020 N = 337
 2021 N = 183
 2022 N = 463

The output shows that all beta credibility intervals cross 0.

Situational Perception Score

Bayesian regression for how positive are students perceptions of open and reproducible research. The scores are between -2 and 2 but will be changed by adding 2.001. This way the scale will be from 0 to 4.

The explainer variables are:

* Whether the participant was in their first or second year of study * The year the survey was conducted - 2020, 2021 and 2022

```
## Inference for Stan model: anon_model.
## 4 chains, each with iter=10000; warmup=5000; thin=1;
## post-warmup draws per chain=5000, total post-warmup draws=20000.
##
##           mean se_mean   sd  2.5%  25%   50%   75%  97.5% n_eff Rhat
## mu          0.40    0.00 0.06  0.27  0.36  0.40  0.44  0.52 12100   1
## sigma        0.49    0.00 0.01  0.47  0.48  0.49  0.50  0.52 23172   1
## sigma_a      0.14    0.00 0.06  0.07  0.10  0.13  0.16  0.28  9559   1
## alpha[1]     0.30    0.00 0.03  0.23  0.27  0.30  0.32  0.36 13589   1
## alpha[2]     0.36    0.00 0.09  0.17  0.30  0.36  0.42  0.53 20821   1
## alpha[3]     0.42    0.00 0.07  0.28  0.37  0.42  0.47  0.57 19871   1
## alpha[4]     0.51    0.00 0.09  0.35  0.45  0.51  0.57  0.69 16972   1
## alpha[5]     0.27    0.00 0.10  0.07  0.21  0.27  0.34  0.45 16396   1
## alpha[6]     0.53    0.00 0.06  0.42  0.49  0.53  0.57  0.64 11079   1
## alpha[7]     0.42    0.00 0.12  0.19  0.35  0.42  0.50  0.65 17366   1
## alpha[8]     0.47    0.00 0.05  0.37  0.44  0.47  0.50  0.57 11374   1
## alpha[9]     0.30    0.00 0.06  0.17  0.26  0.30  0.34  0.42 12236   1
## beta1        0.10    0.00 0.03  0.04  0.08  0.10  0.12  0.17 17782   1
## beta2       -0.08    0.00 0.05 -0.18 -0.11 -0.08 -0.04  0.02 14303   1
## beta3       -0.05    0.00 0.05 -0.14 -0.08 -0.05 -0.01  0.05 10374   1
## lp__        218.63    0.04 3.02 211.78 216.82 218.97 220.80 223.49  7381   1
##
## Samples were drawn using NUTS(diag_e) at Sat Feb 21 11:49:04 2026.
## For each parameter, n_eff is a crude measure of effective sample size,
## and Rhat is the potential scale reduction factor on split chains (at
## convergence, Rhat=1).
```

Number of participants in the analysis

Total N = 976

Year 1 N = 565

Year 2 N = 411

2020 N = 330

2021 N = 183

2022 N = 463

The output shows that beta credibility intervals do not cross 0 only for the Beta 1.

2.5% = 0.0358744

97.5% = 0.1656467

The interpretation is that students across all waves, who were in their second year of study were more likely to have more positive situational perceptions of open science practices compared to students in the first year of study.